

PowerRanger

A software control plane for grid-responsive datacenters

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The Problem

AI trains in **lockstep**, and the grid pays for the rhythm.

Synchronized GPUs swing datacenter power by

hundreds of megawatts in an instant, and those oscillations can **stress and fatigue grid generators**.

We must bound both the **magnitude** and the **rate of change** of datacenter load.

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The Proposal · PowerRanger

PowerRanger holds datacenter power in a **range the grid can move**.

Four per-node APIs define it — **CPU and GPU versions** of each:

set_power_upper_bound	Ceiling (MW) — power may never draw above it
set_power_lower_bound	Floor (MW) — power may never drop below it
set_ramp_up_rate	Climb limit (MW/s) — how fast the band rises
set_ramp_down_rate	Shed limit (MW/s) — how fast power falls

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🔌 PowerRanger in Action

⚡ Grid DR Signal

`set_power_upper_bound(210)`

`set_power_lower_bound(144)`

`set_ramp_up_rate(5.0)`

`set_ramp_down_rate(5.0)`

→ *set the power range*

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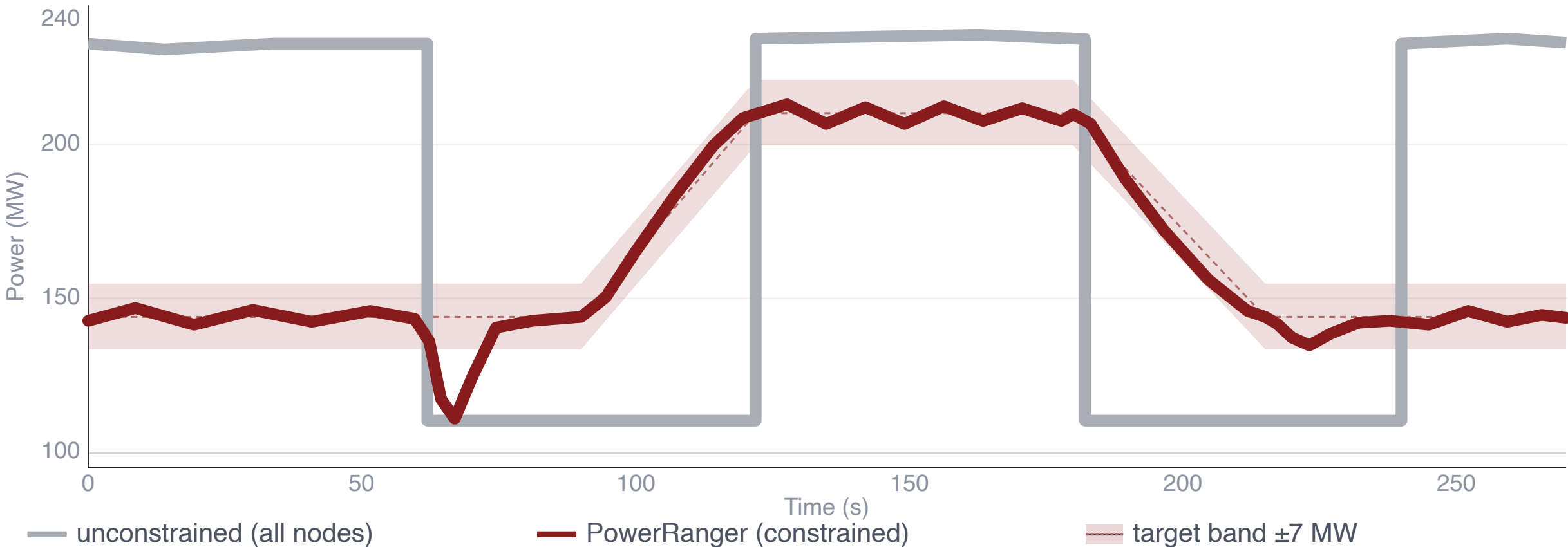
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